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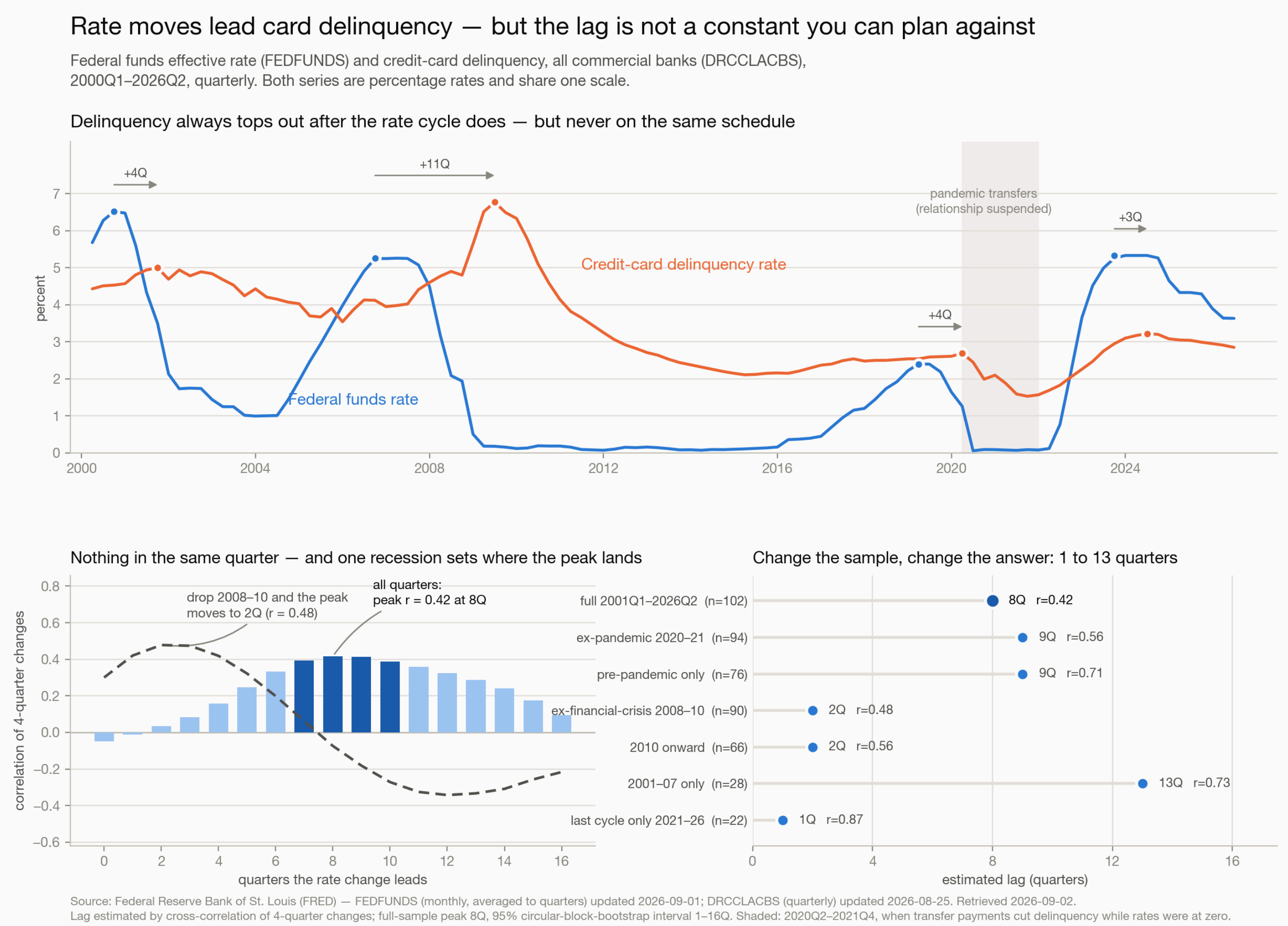
Rate moves lead card delinquency — but the lag is not a constant



Federal funds rate and credit-card delinquency rate, 2000Q1–2026Q2, with the estimated lag between them and how much that estimate moves when the sample changes

The finding

Rate changes lead credit-card delinquency changes. That much is solid: the same-quarter correlation between the two is **–0.05** — effectively nothing — and the correlation only becomes positive once the rate series is pushed several quarters into the past. Nothing about today's rate tells you anything about today's delinquency.

How far it leads is the part that does not hold still. Over the full 2000–2026 sample the correlation peaks at **8 quarters (r = 0.42)**. Drop the 2008–10 financial-crisis quarters and the peak moves to **2**

quarters (r = 0.48). Restrict to 2010 onward and it is 2 quarters again; restrict to 2001–07 and it is 13. The headline "about two years" is carried by one episode.

The cycle-by-cycle record says the same thing. The gap between the quarter the federal funds rate reached its cycle top and the quarter delinquency peaked was **+4, +11, +4 and +3 quarters** across the four cycles since 2000. The +11 is 2006Q3 → 2009Q2: the one cycle in which a deep recession arrived between the two events. In the three cycles without one, delinquency turned within three to four quarters.

The honest reading is that the rate cycle is a **regime signal with a wide and condition-dependent window**, not a lag constant. What appears to set the length of the window is whether a recession intervenes — a hypothesis this two-series analysis suggests but cannot itself test.

Results

Data. FRED series `FEDFUNDS` (Federal Funds Effective Rate, monthly, 320 observations) and `DRCCLACBS` (Delinquency Rate on Credit Card Loans, All Commercial Banks, quarterly, 106 observations), both from 2000-01-01. Retrieved 2026-09-02; FRED last updated FEDFUNDS 2026-09-01 and DRCCLACBS 2026-08-25. Coverage runs to **2026Q2** — the delinquency series publishes about a quarter behind, and that is the binding constraint on how current this can be.

Cleaning and alignment. FEDFUNDS was averaged from months into quarters (the mean, not the period-end value — a quarter's policy rate is what was in force across it). The two were inner-joined on quarter, giving **106 complete quarters, 2000Q1–2026Q2, with no missing values, no gaps and no duplicated periods.** Both series are percentage rates, so they share one axis in the chart and a difference in either is a percentage point.

Analysis is on **4-quarter changes** of each series (102 usable quarters, 2001Q1–2026Q2) rather than levels — two trending levels correlate for reasons that have nothing to do with transmission. The measured quarter-of-year pattern in the delinquency series is small (mean quarter-on-quarter change ranges from +0.01pp in Q1 to −0.04pp in Q3), so the 4-quarter difference is there to damp noise and match the annual cadence of a rate cycle, not to remove a large seasonal.

Lag estimates by sample:

SAMPLE	N	PEAK LAG	R AT PEAK
Full, 2001Q1–2026Q2	102	8 quarters	0.42
Excluding pandemic quarters (2020–21)	94	9 quarters	0.56
Pre-pandemic only	76	9 quarters	0.71

SAMPLE	N	PEAK LAG	R AT PEAK
Excluding financial crisis (2008–10)	90	**2 quarters**	0.48
2010 onward	66	2 quarters	0.56
2001–07 only	28	13 quarters	0.73
Last cycle only, 2021–26	22	1 quarter	0.87

The last two rows are short samples of two trending series and should not be read as estimates; they are in the table to show the range the method spans, not to be believed.

Precision. Even within the full sample the peak is broad and weakly identified. Lags of 7 to 10 quarters are statistically indistinguishable from the 8-quarter peak. A circular block bootstrap (2,000 resamples, 24-quarter blocks, chosen longer than the longest lag scanned so long-lag correlations are not computed across a seam) puts the 95% interval on the peak at **1 to 16 quarters** — the entire range scanned. Only 15% of resamples peak at exactly 8 quarters, and 40% land within one quarter of it. With 26 years of data containing four rate cycles, the effective sample is four events, not 102 quarters.

Effect size. Fitting delinquency change on rate change 8 quarters earlier gives **0.18pp of delinquency per 1pp of rate move** (Newey-West standard error 0.09, $p = 0.047$, $R^2 = 0.17$, $n = 94$). The direction is right and the magnitude is plausible, but R^2 of 0.17 means the rate path leaves five-sixths of the variation in delinquency unexplained.

Direction of causation is not established by this data. Granger tests on quarter-on-quarter changes fail to reject the null at every lag order tried (4, 6, 8, 10, 12 quarters; best $p = 0.13$ at 12 lags). The reverse direction is firmly rejected as a story ($p \geq 0.73$ at every order), so nothing here suggests delinquency leads rates — but "rates Granger-cause delinquency" is not something this sample supports at conventional significance.

Where the relationship is known to break. 2020Q2–2021Q4 is shaded in the chart: transfer payments drove delinquency to a series low of 1.53% (2021Q3) while rates were at zero, the opposite of the historical pattern. It is excluded in the robustness cuts rather than modelled. In the current cycle, delinquency began rising a quarter *before* the Fed began hiking (trough 2021Q3, liftoff 2021Q4), so that rise is at least partly normalisation from suppressed pandemic levels and should not be attributed to rates wholesale.

Where the cycle stands now. Rates topped at 5.33% in 2023Q3 and are at 3.63% as of 2026Q2. Delinquency peaked at 3.22% in 2024Q2 and has fallen to 2.85%. Both are in the post-peak phase, consistent with the historical pattern of a three-to-four-quarter gap outside a recession.

What follows for a risk team

1. Do not hard-code a lag. Any model carrying a fixed "delinquency responds to rates with an N-quarter lag" is fitting one sample. N is 2 or 8 or 13 depending on which years are in the window, and the confidence interval on the full-sample estimate spans everything from one quarter to four years. If a lag constant is already embedded in a scorecard or a provisioning model, that is worth finding and re-testing on a rolling window.

2. Use the rate cycle as a watch trigger, not an input. The defensible use is: when a tightening cycle tops out, open a heightened-monitoring window on the card book and keep it open for roughly one to three years. That is coarse, but it is what the data supports, and it is more useful than a precise number that is wrong.

3. The variable that seems to set the timing is not in this analysis. The one cycle with an 11-quarter gap is the one with a deep recession in it. The next step is to bring the labour-market series in — unemployment (`UNRATE`) is the obvious candidate — and test whether it, rather than the rate path, is what carries the delinquency peak. If it does, the early-warning indicator a risk team should actually watch is employment, with rates as context.

4. Resist the forecast. Applying the fitted 8-quarter relation to rate cuts already in the data implies delinquency roughly 0.1 to 0.3pp lower through 2027Q2. The 80% prediction interval on that is about ±0.85pp — wider than the entire move being predicted, and wide enough to include a material *increase*. The relationship is real enough to inform when to look; it is not precise enough to put a number in a provision.

5. Refresh on the delinquency calendar, not the rate calendar. FEDFUNDS updates monthly; DRCCLACBS lands roughly a quarter in arrears. Any monitoring built on this pair moves at the slower series' pace, and the current reading will always be about a quarter stale.

Source: Federal Reserve Bank of St. Louis (FRED) — `FEDFUNDS` *(monthly, averaged to quarters) last updated 2026-09-01;* `DRCCLACBS` *(quarterly) last updated 2026-08-25. Retrieved 2026-09-02 via the workspace* `fetch-fred` *skill. Analysis in* `analysis.py`*, chart in* `chart.py`*, full numeric output in* `data/analysis.json`.
